

Application Form - PN-IV-RU-SC-TE-2026-2

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Title:

**Magnetically assembled plasmonic hotspots for
pre-concentrated SERS detection of pollutants in flow**

Acronym: MAGNIFY

Project description

B.1. Project Director's Performance

General competence and baseline scientific skillset

Dr. Fran Nekvapil (the PD) did his Master programme in marine biotechnology field at the University of Dubrovnik (Croatia). Upon graduation, he moved to the Faculty of Physics (Babeş-Bolyai University - BBU (Romania), firstly as an Erasmus+ exchange student, and afterwards as a regular Ph.D. student. He defended his thesis in October 2020 titled “Raman Spectroscopy Techniques and Applications to Carotenoids Research in Marine Living Resources and to the Blue Bioeconomy” with excellent mark. Currently, Fran is a full-time researcher at the National Institute for Research and Development of Isotopic and Molecular Technologies (INCDTIM) - host institution. He is a part of the [NanoPhotoniX](#) group, investigating spectroscopy fundamentals and developing new optical sensors.

Raman methodology development: Dr. Nekvapil specializes in applied Raman spectroscopy (RS), including multi-laser Raman, Resonance Raman, Surface-Enhanced Raman Spectroscopy (SERS), and FT-Raman. His work is complemented by analytical techniques such as UV–Vis absorption, FT-IR (transmission and ATR), electron microscopy, XRD, magnetometry, and N₂ physisorption (BET). Although he does not routinely perform non-spectroscopic measurements, he is proficient in interpreting and integrating these data into comprehensive sample characterization. He has extensive expertise in spectral pattern recognition, as well as the in-depth analysis and deconvolution of Raman spectra.

The PD has created a companion website to this project proposal, at the Github repository, where he has uploaded the qualitative achievements and preprints of manuscripts in progress, for Evaluator's reference. The site can be accessed at <https://frannekvapil.github.io/TE2026/> [1].

SERS molecular tracking: Dr. Nekvapil recently began working with self-assembled film-over-nanospheres (FoN) plasmonic substrates. He investigated the influence of focal position relative to gold-coated FoN (AuFoN) substrates [2] and demonstrated nanomolar-level quantification of the natural antioxidant astaxanthin by combining Resonance Raman spectroscopy with silver-coated FoN (AgFoN) substrates [3]. He is currently preparing a manuscript comparing resonance and non-resonance excitation for advanced probing of astaxanthin adsorbed on AgFoN. Previously, he worked extensively with colloidal silver nanoparticles (AgNPs), publishing several studies on *in situ* tracking of biological compounds [4,5], and has also performed measurements using proprietary SERS substrates, including Klarite and SERSitive.

Adsorbent materials: During his 2022–2023 postdoctoral project Shell-Pol-Ads ([PN-III-P1-1.1-PD-2021-0477](#)) [6], Dr. Fran Nekvapil investigated the recycling of nanoporous crab shells, a food-processing waste, as bio-based adsorbent materials within the framework of Blue biotechnology. The project explored physical and chemical approaches to increase pore surface area and resulted in two research articles: one on the comparative characterization of different shell types and another on the adsorption of the pesticide acetamiprid using blue crab (*Callinectes sapidus*) shells [7]. The work also led to a patent for a method of producing adsorbent materials with a pore surface area of 200–300 m² g⁻¹ [8].

Ferrites characterization: Upon joining INCDTIM in 2018, Dr. Nekvapil began working with metallic nanoparticles (NPs) through the SANOMAT project ([PN-III-P1-1.2-PCCDI-2017-0062](#)) [9], which explored magnetic ferrite nanoparticles for medical applications. Using Raman spectroscopy, he established a correlation between the cation-vacancy band and saturation magnetization in Mn–Zn ferrites [10], a study that has since attracted considerable scientific interest – 45 citations. He later contributed to the ZnOFluoSERS project ([PN-III-P1-1.1-TE-2019-1141](#)) [11], investigating the plasmonic properties of ZnO nanoparticles and reporting excitation-dependent band broadening in nano ZnO synthesized by different methods [12].

Lattice transitions under annealing up to 1100 °C have also been comparatively investigated by RS and XRD on Mn-Zn ferrite NPs, and the analytical advantages and shortfalls of either method were reported [13]. The PD also characterized the structure of substituted Co-Mn and Mn-Co ferrite NPs. Occasional RS

measurements were also done on Zn-ferrites and Ni-ferrites. All of these measurements and deconvolution procedures have yielded a deep understanding of Raman signal of ferrites.

Relevant preliminary data

Recently, dr. Nekvapil has embarked on testing the wavelength-dependent Raman signal response of Co-ferrite NPs. The motivation for this study came from the observation that Raman signal of Co-, Zn- and Mn-ferrites does not tightly correspond to their stoichiometry when excited with the 632,8 nm laser. Hence, this study in progress is directly linked to the current proposal components, as it tested the behaviour of Co-ferrite core under 532 and 632.8 nm excitation. Overall, Co-ferrite NPs were irradiated with 4 different laser lines spanning from UV to NIR range, and the PD observed differential enhancement of the Co-sublattice. RS measurements were processed, while theoretical background is currently being refined.

Most important contribution to the research domain

Dr. Nekvapil's most important contribution to the field is the development and formalization of an interpretation framework for structural characterization of ferrites NPs via Raman signal.

The framework currently exists as an [internal „Method file”](#) of the INCDTIM, periodically updated as the new data become available to the PD. It is intended to speed-up the learning curve for new spectroscopists entering the nanomaterials engineering domain through providing baseline structure-signal relationships. Dr. Nekvapil intends to publish this issue as a review-type article in near future.

Management and leadership competence

Dr. Nekvapil first led the MARIFERT project ([GTC 31992/02.06.2020](#)) [14], an internal BBU grant that nevertheless involved an evaluated research plan and coordination of an informal research team [13]. He subsequently participated in a programme ([POCU/380/6/13/123886](#)) [15] focused on mentoring of researchers by private business actors. At the same time, he led the nationally funded postdoctoral project Shell-Pol-Ads ([PD 46/2022](#)) [6] evaluated with „[excellent](#)” mark upon completion, carried out under the supervision of a scientific mentor. Through these projects, he gained experience in research leadership, project management, and team coordination, providing a solid foundation for leading a small team within the proposed project.

B.2. Research Project

B.2.1. Ambition and feasibility

Scientific scope and aim

Dynamic colloidal SERS substrates exploit nanoparticle motion, aggregation, or tunable plasmonic coupling to generate transient hotspots that enable highly sensitive and time-resolved measurements. However, their principal limitation remains poor colloidal stability and reproducibility. Because hotspot formation depends on stochastic nanoparticle interactions and Brownian motion, signal intensity can fluctuate significantly over time and between measurements, complicating quantitative analysis [16,17]. The two major challenges of colloidal systems are aggregation (often irreversible) due to the ionic strength, pH and temperature of the matrix [16,18], and long-term stability degraded by potential surface oxidation and analyte fouling [18], both of which lead to inconsistent enhancement factors and inter-laboratory comparability. Despite major advances in dynamic SERS sensitivity and temporal resolution, the field still requires robust control of hotspot dynamics, improved colloidal stability, and standardized substrate architectures before reliable quantitative and routine real-world deployment can be achieved.

Current microfluidic-SERS systems are predominantly built around plasmonic nanostructures integrated within microchannels to generate reproducible electromagnetic hot spots for Raman enhancement. Common substrate architectures employ immobilized gold or silver nanoparticles, nanorods, nanopillars, nanoarrays, or roughened metallic films patterned directly onto channel walls or embedded as sensing zones [19]. Two main

design strategies are used. The first relies on static substrates, where nanostructured metallic surfaces are fixed inside continuous-flow, microarray, or gradient channels, enabling controlled analyte transport over reproducible SERS-active regions. The second uses dynamic substrates, where nanoparticle probes are suspended in droplets or flowing streams and assembled during analysis to create transient hot spots [20]. Recent designs extend beyond conventional channels, including droplet-based, digital, lateral-flow, and microarray-integrated devices. These aim to improve analyte–substrate interactions, reduce sample consumption, and increase throughput [20,21].

Overall, the prevailing trend is toward highly integrated lab-on-a-chip platforms that combine precise microfluidic control with engineered plasmonic nanostructures to maximize signal enhancement, reproducibility, and diagnostic performance [19].

Magneto-plasmonic nanomaterials are attractive SERS substrates because they combine plasmonic enhancement with magnetic manipulation, enabling analyte preconcentration, substrate recovery, and active control of nanoparticle assemblies. However, realizing strong magnetic and plasmonic performance simultaneously remains a central materials-design challenge. Magnetic components often introduce optical losses that weaken plasmonic resonances, while highly plasmonic noble metals typically exhibit weak magneto-optical responses. As a result, engineering architectures that preserve both functionalities requires careful control of composition, geometry, interfaces, and nanoscale coupling [22]. For SERS applications, reproducible hotspot formation and structural uniformity remain significant knowledge gaps. At present level of understanding, hybrid core–shell and composite nanoparticles frequently suffer from variations in morphology, interparticle spacing, and magnetic self-assembly behavior [23,24]. Progress in magneto-plasmonic SERS depends on achieving stable, reproducible, and magnetically controllable hotspot architectures while maintaining strong plasmonic enhancement and practical manufacturability.

The materials of choice in this project are the cobalt ferrite nanoparticle cores, covered with gold (AuCo-ferrite) and silver (AgCo-ferrite) shells. The two abbreviations are used interchangeably unless specified otherwise. Co-ferrite was chosen here primarily due to its good chemical stability and for continuity with PD's preliminary work. This ferrite exhibits excellent magneto-optical (MO) properties making it suitable in broader sense for incorporation into miniaturized non-reciprocal optical devices, data storage and spectroscopy elements, in addition to its traditional uses medicine and environmental pollution management. Au and Ag nanostructures are not commonly magnetic, however, they exhibit high localized surface plasmon resonance (LSPR) activity when excited into their resonance band, making them excellent SERS-active components. Although we are not targeting specifically the MO properties of Co-ferrite, there is evidence that interfaces between magnetic and plasmonic nanomaterials cause them to influence each other's properties in ways not yet fully understood [25].

Assessment and monitoring of environmental pollution, both in natural waterways and in industrial effluents is required by law around the World. Dyes are common environmental pollutants generated from textile, chemical, or food industries, with global production estimated at more than 7 million tons annually [26]. Azo dyes constitute the largest class of synthetic dyes currently in use. When discharged untreated, dye-laden effluents substantially elevate biological and chemical oxygen demand, disturbing the balance of the aquatic ecosystems [27]. It is imperative to increase focus on developing environmentally friendly adsorbents with enhanced removal efficiencies for water treatment applications. In particular, magnetic nanoparticle-based adsorbents have emerged as promising candidates because they combine high adsorption capacities with simple and efficient recovery from aqueous media through magnetic separation [28].

One of the driving long-term prospects behind understanding the mechanisms of transient hotspot assembly by AuCo-ferrite is development of capabilities of assessing pollutants not inherently optically active by spectroscopy techniques. This would present several advantages as spectroscopy sensors are generally more portable, field-deployable and resource-efficient than lab based instrumentation (e.g. HPLC systems).

Project scope

We will investigate how dynamic magnetic assembly of magnetoplasmonic NPs controls hotspot density and SERS enhancement under flow conditions. We will further demonstrate the generated knowledge through a custom-designed demonstrator microfluidic platform that may serve as sensing unit for perspective environmental monitoring devices.

The platform will be validated through adsorption of azo-dyes. Dyes are optically active compounds, allowing for the use of UV-Vis spectroscopy to support our SERS measurements, which is well within the capabilities of our infrastructure. Additionally, we will assess the detection capacity of our demonstrator platform for two other environmental pollutants: the pesticide thiabendazole (TBZ) and the anticancer drug 5-fluorouracil (5-FU) which are known to exhibit a SERS signal on Au and Ag plasmonic substrates, respectively [29,30].

The proposal identifies several research questions, such as (Q1) How do the MO Co-ferrite interact with the covering Au/Ag plasmonic nanostructures? (Q2) Is it possible to treat a large volume of effluent with the AuCo-ferrite and consequently use an external magnetic field to concentrate and analyze the adsorbed matter? (Q3) How do the magnetically assembled SERS hotspots behave under flow?

To counter these questions, we come with several hypotheses: (H1) There exists an optimal Au/Ag coverage which yields maximum SERS enhancement while still maintaining magnetic manipulability, (H2) The action of external magnetic field brings AuCo-ferrite NPs into direct contact within a defined area, enabling assembly of numerous transient hotspots, and (H3) Magnetically induced assembly of hotspots enables greater SERS detection rate than conventional static substrates and randomly dispersed colloidal noble metal NPs. Hence, the project's objectives are as follows:

- **O1: Optimization of AuCo-ferrite magnetic and plasmonic properties (H1); establish a synthesis method that will produce AuCo-ferrite within defined parameter range.**
- **O2: Study magnetically assembled hotspots in clusters and in flow; create a demonstrative microfluidics chip to facilitate study in flow conditions (H2).**
- **O3: Validate and establish the performance of the magnetically assembled hotspots for tracking of common environmental pollutants within the microfluidics platform (H3).**

Novelty

The project will generate novel insights into assembly of hotspots in the broader context of magnetoplasmonic nanomaterials engineering, which is a young field with considerable knowledge gaps and optimization potential. Furthermore, we re-invent how a sensing platform behaves (Table 1).

Table 1. Advancements proposed in this project relative to current state-of-art.

<u>Current materials and common limitations</u>		<u>Our advancement</u>
Magnetoplasmonic particles – passive characterization	→	Real-time magnetic assembly
SERS substrates – static or randomly dispersed hotspots	→	Dynamically controlled hotspots
Microfluidic SERS – immobilized substrates	→	Mobile magnetic substrate
Ferrite-Au particles – sensing only	→	Flow-controlled sensing platform

Research methodology overview

Noble metal covered Co-ferrite NPs will be synthesized through chemical methods. Core size and shell thickness will be changed through variation of synthesis parameters. Magnetic and LSPR properties will be evaluated for each batch, until NPs with balanced magneto-plasmonic properties are obtained. Figure 1. Shows schematically the workflow through the objectives.

Finite-Difference Time Domain (FDTD) modelling will be employed throughout all stages of the workflow. Distribution of electromagnetic field and hotspots will be studied by modeling the AuCo-Ferrite NPs in

isolation, clusters and flow conditions. The custom designed 3E-printable demonstrator microfluidics chip will allow the validation of AuCo-ferrite NPs response in experimental setups with different adsorbed analytes.

Finally, we shall study the adsorption capacity, and use the acquired insights into behaviour of our magneto-plasmonic NPs to demonstrate pre-concentration and tracking of environmentally relevant analytes, such as pollutants azo-dyes, TBZ, and a drug 5-FU.

Infrastructure and feasibility

This project will be conducted at INCDTIM. The PD and team member I.C. work in the Nanophotonix team. Relevant facilities include the Micro/Nano Fabrication Laboratory equipped with self-assembly and sputter-coating devices needed for AuFoN fabrication (<https://eertis.eu/erlb-2531-323h-74565>). Available facilities further include a “clean room” equipped with an E-beam film deposition system (<https://eertis.eu/ereq-2400-011n-4779>) The Vibrational spectroscopy lab (<https://eertis.eu/erlb-2531-323r-74428>), is equipped with JASCO FT/IR-4600 Spectrometer and the μ Sight-2100 module of the Technospex system (<https://eertis.eu/erlb-2531-323g-74788>). Furthermore, the same lab is equipped with a Portable i-Raman Spectrometer BW-TEK 532S (532 nm excitation) with an optical fiber connection for a trigger probe or a digital microscope, and a Technospex μ Raman system (632.8 nm excitation) (that shall be used for validation of SERS performance of the AuCo-ferrite NPs.

Both the synthesis of the nanomaterial and adsorption testing will use installations of the Chemical synthesis laboratory (<https://eertis.eu/erlb-2531-323e-74479>). Synthesis and initial characterization of nanomaterials: magnetic stirrer; ovens; glove box for electrode preparation; centrifuge; „furnaces (1600°C; Adsorption tests: T90+ UV-Vis spectrophotometer (<https://eertis.eu/ereq-2500-013v-3285>)§ orbital shakers and stirring platess permanent neodymium magnets.

Other labs: XPS(UPS)- SPECS (<https://eertis.eu/erlb-2531-323k-74511>); XRD- Rigaku Smart Lab (<https://eertis.eu/erlb-2531-323j-74491>); VSM (Cryogenic) and SQUID Magnetometer (<https://eertis.eu/erlb-2531-323e-74524>); PL (JaskoFP-6500) –luminescence properties and UV-Vis spectromether (JASCO V570 UV– VIS-NIR with absolute reflectivity measurement JASCO ARN-475).

There are several 3D printers at INCDTIM that use different filament materials, such as the MarkForge OnyxPro (Gen 2) (<https://eertis.eu/erlb-2531-323j-74689>) and a Creality desktop 3D printer.

In addition to experimental measurements, predictive modelling (member E.V.) will be achieved through the use of High-performance Computing infrastructure at UBB (<https://eertis.eu/ereq-2300-005f-7530>) and Computational Science and AI-enhanced Research Development at INCDTIM (<https://eertis.eu/erlb-2531-323f-74797>).

Adjacent to both departments is the Integrated Electron Microscopy Laboratory (, equipped with TEM/STEM (Hitachi SU8230 with EDX /EELS system, which will be used for probing the morphology and coverage of Co-ferrite NPs with Au/Ag films (<https://eertis.eu/erlb-2531-323k-74599>). The PD, I.C. and E.V. also have access to top-notch Raman spectrometer, Renishaw InVia Reflex (<https://eertis.eu/ereq-2200-002t-6679>) in case additional level of precision will be needed in validation and optimization activities.

B.2.2. Impact

MAGNIFY will deliver (1) reproducible synthesis protocols for magneto-plasmonic AuCo-ferrite and AgCo-ferrite NPs, (2) new knowledge on the formation and behavior of transient SERS hotspots under flow conditions, and (3) a real-world demonstration of magnetic pollutant pre-concentration and SERS detection. The generated knowledge will be disseminated through research articles, conferences, and data-rich website.

The experimental and theoretical datasets generated will support further advances in magnetoplasmonic materials and SERS sensing. In addition to successful outcomes, the project website will document relevant experimental trials, providing the scientific community with a broader understanding of material behavior. The

project will contribute to environmental engineering by validating a novel optical sensing concept applicable to contaminants that are not inherently optically active.

MAGNIFY will also establish a new research team with expertise in the fundamental and applied study, innovation, and optimization of magnetoplasmonic nanoparticles, strengthening capacity for participation in future national and international projects.

The team will complement its portfolio of dynamic research groups at the host institution, and create opportunities for technology valorization, including intellectual property related to the microfluidic sensing platform. Team members will benefit from interdisciplinary collaboration and international networking, and together with the host institution achieve increased visibility through conference participations and scientific publications. The PD will further develop leadership experience and strengthen the capacity to pursue larger collaborative funding opportunities.

B.2.3. Work plan and schedule

Structure of the research team

Total 5 team members, 27 person-months/year (54 PM total effort), total net budgeted personnel expenditure (BPE) = 412 000 RON (cca. 82 400 EUR)

Dr. Fran Nekvapil (physics, spectroscopy), the PD ; <https://ro.itim-cj.ro/portfolio/nekvapil-fran/>

$\frac{3}{4}$ of the full norm (126 hr/month); The responsibilities of dr. Nekvapil include idea leadership, conceptualization of work tasks and extensive literature reviewing to keep up-to-date with current scientific context. The PD will be involved in all Work Packages, but especially WP3 and WP4 relating to design and validation of the microfluidics platform and project management, respectively. In addition to natural convergence of his work to magneto-plasmonic nanomaterials, tasks are conducive the overall aspirations of the PD to become a interdisciplinary coordinator.

Drd. Ioana Cardan (physics, spectroscopy) ; <https://ro.itim-cj.ro/portfolio/ing-cardan-ioana/>

$\frac{1}{2}$ of the full work norm (84 hr/month); Ioana is a rising spectroscopy expert and spectral data processor. At the submission date, she is finalizing her doctoral thesis and is poised to defend it by the implementation date, hence she will effectively be a post-doc. She is currently working on SERS substrates fabrication and ultra-low concentration molecular tracking of nanoplastics. Ioana also has basic knowledge of SEM imaging of plasmonic substrates. Her principal responsibilities will include testing of plasmonic properties of newly synthesized AuCo-ferrite, and SERS validation, evaluation and benchmarking of the AuCo-ferrite in clusters and flow (WP1 and WP2).. Participation on this project will yield her novel experiences with SERS substrates and their validation in a context not considered before in her career.

Drd. Ecaterina Vlaico - Ph.d. student (spectroscopy, computational physics)

$\frac{1}{2}$ of the full work norm (84 hr/month); Ecaterina is at the beginning of her Ph.D. cycle. The PD knows her for several years now from the lab team at BBU, and he knows that she is ready to learn and overcome any obstacle with confidence. Her main tasks will comprise modelling of electromagnetic field of the AuCo-ferrite in static and flow conditions (WP2). She will also assist in dissemination and data management. She will benefit from additional problem-solving experience and different aspects and possibly innovative modelling, which is considerably helpful in her current stage.

Drd. Ana Varadi (chemistry, nanomaterials engineering) ; <https://ro.itim-cj.ro/portfolio/varadi-ana/>

$\frac{1}{4}$ of the full work norm (42 hr/month); Ana is a Ph.D. student at the moment of submission, but is likely to have defended the thesis by the project start. She is a chemist, specialized in metallic nanomaterials engineering. Her main responsibility will be optimizing the synthesis methods for AuCo-ferrite and AgCo-

ferrite and further supply the optimized substrate (WP1). This will yield her new experience in the nanostructure synthesis and expand her portfolio nanomaterial types worked with.

Dr. Iolanda-Veronica Ganea (environmental engineering, water decontamination)

<https://ro.itim-cj.ro/portfolio/ing-ganea-iolanda-veronica/>

¼ of the full work norm (42 hr/month); The PD has worked with Dr. Ganea on separate occasions, and knows that she is a capable and dependable water quality researcher, including organic and inorganic pollutants, heavy metals, and mineral content. She also did a lot of innovative work on methods for water decontamination. Her tasks here will involve adsorption capacity and kinetics testing of the AuCo-ferrite NPs in static and flow conditions (WP3). This project will yield her novel networking opportunities and insights into adsorption mechanics between metals and dyes, and spectroscopy aspects of pollution monitoring.

WP1: Synthesis and characterization of AuCo-ferrite and AgCo-ferrite NPs (resp. A.V., the PD, I.C.)

- **Activity 1.1.:** Synthesize series of Au/AgCo-ferrite NPs with incrementally varying magnetic core size and plasmonic coating thickness;
- **Activity 1.2.:** Characterize the synthesized Au/AgCo-ferrite magnetic and plasmonic properties and their deviation from desired properties -feedback loop into A1.1.

Milestone 1: Reproducible synthesis method for optimized AuCo-ferrite NPs with bracketed variation of properties, with characterized and cross-validated SERS performance.

Technical: starting point are established Co-ferrite synthesis methods: wet chemical methods (co-precipitation, solvothermal or hydrothermal). Variation of experimental parameters controls the size and overall properties of the nanocomposite system, while in control favors the increase of the dispersability. Noble metals (Au/Ag) will be dissolved by hydrazine/citrate reduction of corresponding inorganic salts and shells will be deposited using different interactions (covalent, non-covalent, electrostatic, polymeric wrapping) of the original components).

Since this is a frontier study, it is not possible to predict the exact NPs parameter, however we can suggest feasible starting points. Co-ferrite cores of around 50 nm may provide good magnetic manipulability, while Au/Ag shell of 5 – 20 nm thickness should give LSPR in 550 – 600 and 400 – 450 nm range, respectively.

Magnetic moment measured by VSM will be important, but we will also monitor the shell thickness uniformity, imaged by SEM, as it will directly influence the electric field distribution.

WP2: Electromagnetic modelling and assessment of SERS capacity of AuFoN (resp.: the PD, I.C., E.V.)

- **Activity 2.1.:** Use FDTD predictive electromagnetic simulations to model hotspots in cluster assemblies and flow conditions; as a function of interparticle distances, transient hotspots distribution in the magnetic assembly and overall correlations with experimental observations;
- **Activity 2.2.:** Validate the SERS performance of optimized magneto-plasmonic NPs against a common, FoN substrate: using the same noble metal cover and resonant laser line (532, 632.8 nm);
- **Activity 2.3.:** Validate and establish the performance parameters of the microfluidics chip, such as signal loss, analysis volume limitation and overall reproducibility of the signal acquisition and intensity using a standard analyte; benchmark against Au/AgFoN.

Modelling: Year 1: focus on modelling the local electromagnetic field distribution as a function of NP size and morphology, as well as investigating the hotspot formation within NP cluster assemblies. Year 2: focus

on the electromagnetic field distribution and intensity as a function of interparticle distance, together with the spatial and temporal evolution of electromagnetic hotspots in flow conditions.

Validation and evaluation: This cross-cutting work package supports all project objectives (Figure 2a). FDTD modelling will be used to predict electromagnetic enhancement and will be validated through experimental determination of SERS enhancement factors, followed by comparison of the two approaches. Validation of AuCo-ferrite nanoparticles in suspension and under flow conditions will employ reference dyes with well-characterized optical properties, enabling benchmarking against AuFoN substrates. Silver-coated systems will be evaluated using Nile Blue (NB) with 532 nm excitation, while gold-coated systems will be tested with Rhodamine 6G (R6G) under 632.8 nm excitation. For microfluidic experiments, a spectrometer probe will monitor the analysis window during continuous sample in “loop” measurement setting flow. The resulting data will be analysed as a function of flow rate and sample characteristics to establish recommendations for optimal chip operation.

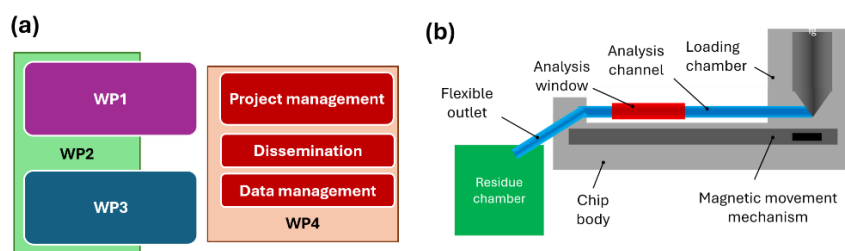


Figure 2. (a) Schematic presentation of relationship between Work Packages; (b) The pre-implementation, unrefined, conceptual design of the microfluidics chip.

WP3: Design of the microfluidics chip and demonstration of magnetic hotspots assembly (resp.: the PD, I.V.G.)

- **Activity 3.1:** Develop the demonstrator microfluidics chip blueprint; optimize the sample suspension movement system; design versatility features for use with microscopes and optic fiber probes;
- **Activity 3.2.:** Assess the efficacy of tracking of two azo-dyes, one compatible with AgCo-ferrite and 532 nm, and one with AuCo-ferrite and 632 nm excitation. Determine LOD and signal reproducibility (consistency of transient hotspot assembly). Optimize analytical procedure and write signal interpretation framework for each.
- **Activity 3.3.:** Assess the capacity of the sensor platform for tracking of TBZ and 5-FU which are not inherently optically active. Optimize analytical procedure and write signal interpretation framework for each.

Milestone 2: Blueprint for the microfluidics chip with semi-automated magnetic mechanism (Figure 2b).

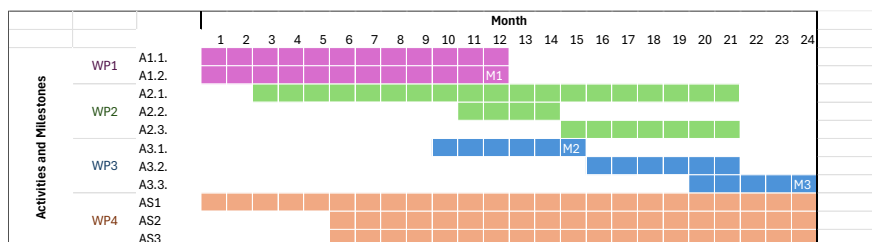
Milestone 3: Real-world demonstrator proved to work; best analytical procedures established

Adsorption equilibrium will be evaluated by exposing a fixed mass of AuCo-ferrite NPs to a series of dye solutions of different initial concentrations, fitted to standard adsorption isotherm models, allowing determination of the maximum adsorption capacity of the AuCo-ferrite NPs and providing insight into whether adsorption occurs via monolayer coverage on homogeneous sites or via a more heterogeneous, multilayer process. Basic effectiveness of SERS detection will be validated through parallel UV-Vis quantification of dye adsorption according to standard literature procedures. SERS enhancement offered by the chip platform will be benchmarked against the homologous results obtainable on other plasmonic substrates. Effectiveness of

this process will be validated through SERS response, against common FoN-type SERS substrate our team is working with on regular basis (objective 3).

WP4: Project management and dissemination (resp.: the PD)

- Activity 4.1. Project management, finances, and reporting
- Activity 4.2. Dissemination
- Activity 4.3. Research data management



Dissemination

Communication: to raise awareness about this project, its funders, and the team an online campaign will be in running on relevant web platforms (social media, professional profiles, institutional websites).

Dissemination: We will target at least 4 international conferences concentrated in the second half of the implementation period. The conferences will be selected by topic area, to allow both the spectroscopy and the nanomaterials engineering components of the team a chance for relevant networking.

Original research articles will be published in Q1 and Q2 ranked journals. We reasonably estimate submission of at least 3 research articles, two of which might be published during the implementation: (i) structure-property relationships of AuCo-ferrite, (ii) behaviour of AuCo-ferrite under flow conditions, and (iii) real-world demonstration of pollutant assessment using our magneto-plasmonic platform.

The implementation will close with a demonstrative workshop with regional invitations, organized at the host institution INCDTIM. The workshop will present the achievements and insights of the project, and aims to showcase the newly formed team and its capabilities.

Exploitation: the microfluidics chip blueprint, with its accessories (with) AuCo-ferrite NPs synthesis method will be patented as the intellectual property of the host institution (INCDTIM), with the patent application submitted to OSIM (Romanian patent office) within the duration of the project.

Open science

Generated datasets and relevant lab notes will be made publicly available through the official project website and permanent online repositories. Scientific articles will be published in Open access regime to promote immediate visibility of the team and make the research available to all stakeholders. The workshop will feature a scientific/informative/educative and a demonstrative session, and be open to any interested person, academic, business or citizen interested in science.

Research data management

Electronically created data (ferrites characterization data, spectral data in ASCII format) will be stored (1) on a dedicated physical drive, (2) on a reputable online repository (e.g. Github, Zenodo or Figshare), and (3) in a

form accessible from the project website. Relevant lab notes will be digitalized periodically. Full transparency of the research data will be respected, and data will be published in standard formats according to the FAIR principle [32]. Microfluidics chip blueprint will be (partially) released after IP stage according to the principle “as open as possible, as closed as necessary” [33].

Gender equality, inclusion and diversity

The proposal implementation will follow the institutional Gender Equality Strategy 2026-2030 of the INCDTIM [31]. The composition of the team results from previous work experience of the PD and personnel available at the host institution. The PD will ensure transparent allocation of tasks based solely on scientific capabilities of either team member, equal conference travel opportunities and objective authorship criteria. Mentorship opportunities will be made available to junior team members. The PD has Croatian ethnicity, one team member has Moldovan ethnicity, and other three are Romanian.

Risk Assessment

Risk	P	I	Sc.	Mitigation
Shell wavelength mismatch: LSPR band shifted away from 632.8 and 532 nm due to MO effect of Co-ferrite core	3	4	12	tune shell thickness during WP1 with FDTD guidance; also use supplementary laser lines available at the Renishaw spectrometer at BBU
Magnetic aggregation decreases reproducibility	5	4	20	AggregationIntroduce dispersants in synthesis; induce turbulence during adsorption trials ; larger microfluidics analysis channel geometry.
Low analyte quantity within hospots due to poor adsorbent capacity of the shell	3	7	21	Switch to analytes containing a thiol group; surface functionalisation to direct adsorption to the metal
Au/Ag shell fails to give sufficient enhancement	3	8	24	Switch to different morphology: Au/Ag islands, nanoflower decoration
Ag shell instability: oxidation in aqueous/flow conditions.	6	5	30	ultrathin Au or other stabilizer overcoat; prioritise the Au and treat Ag as secondary; assess durability of AgCo-ferrite in WP3

P (probability) and I (impact) scale 1 to 10; Sc. (score) scale 1 - 100

B.2.3.1. Budget

Personnel effort allocation by category, 54 PM in total:

- Project director = 18 PM
- Ph.D. Students = 30 PM
- Researcher = 6 PM

Materials and small lab equipment expenses are prioritized in the first year in order to acquire the necessary consumables and small lab devices. Travel expenses are prioritized in the second year as we expect to have a body of obtained results to disseminate at the conferences (CD&E plan).

The most resource-intensive components are A1.1. and A1.2., which will require reagents for nanomaterials synthesis and materials for FoN SERS substrate deposition, hence, these will be prioritized and fully stocked. Dyes and solvents will be required for Ob3. Other equipment includes pipettes, test tubes and sample handling tools.

Small lab devices include e.g. spare hot plate stirrer for the synthesis lab (est. 5000 RON), an equipment of Chemical synthesis lab that will be extensively used, and a long working distance objective (est. 6000 RON) for the Vibrational spectroscopy lab and for better coupling with the microfluidics chip.

Certain team members will need a refreshment of their personal work equipment, hence first-year funds will also be used for acquisition of electronics (e.g. laptops, cca. 5000 RON each). The remaining funds allocated to logistics will cover running needs of laboratories that may become apparent upon the project start to ensure smooth implementation.

This project does not foresee acquisitions of major pieces of equipment. Furthermore, since the host institution, the INCDTIM, is a member of ANELIS association and therefore the authors have Open Access APC waiver, no funds for publication charges have to be budgeted. If this option fails, we will be forced to either negotiate with open access journals or to publish in subscription-based regime.

The PD will dedicate, under normal circumstances, 126 hours/month to the project, and in periods of exceptional workload from other tasks a minimum of 90 hours/month.

Pre-calculation estimate (in lei, per calendar year):

Budget chapter	Year I (lei)	Year II (lei)	Total budget (lei)
Personnel expenses	206000	206000	412000
Logistics expenses	43000	17000	60000
Out of which the value for the equipment expenses	21000	0	21000
Travel expenses	0	31500	31500
Indirect expenses	45600	50900	96500
Total	294600	305400	600000

Pre-calculation estimate (in Euro, at the project level):

Budget chapter	Total budget (euro)
Personnel expenses	82400
Logistics expenses	12000
Out of which the value for the equipment expenses	4200
Travel expenses	6300
Indirect expenses	18300
Total	120000

B.2.4. Research ethics

The project concerns is compliant with the Romanian law on the status of research, development and innovation personell and the ALLEA European Code of Conduct for Research Integrity (2023). There is no conflict of interests arising from this project. and the ALLEA European Code of Conduct for Research Integrity (2023).

¹ <https://www.alleageneralassembly.org/wp-content/uploads/2023/06/European-Code-of-Conduct-Revised-Edition-2023.pdf>

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